Literature Review

GROUP:9,S7C
CHN23CS030 Aleesha M A
CHN23CS101 Jia Chinnu Abraham
CHN23CS102 Joel George Mathew
CHN23CS120 Lois Rose David

SI:NO	PAPER TITLE AND YEAR	METHODOLOGY	EVALUATION METRICS ACCURACY REPORTED	GAPS IDENTIFIED
1	**Multi-Objective EV Charging Station Allocation (2023**	Multi-objective optimization for optimal EV charging station and renewable energy placement using the IEEE 69-bus system	Power loss, voltage profile, operational cost, renewable energy utilization. Accuracy: Not applicable (optimization-based study).	Does not support real-time slot booking, trust evaluation, predictive availability, or user-centric charging services.
2	**Integration of Electric Vehicle into Smart Grid (2024)**	Improved Honey Badger Algorithm (IHBA) for optimizing Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) energy management.	Efficiency, Power Loss, Total Harmonic Distortion (THD)	Does not include charging-slot reservation, real-time charger availability, navigation, or prepaid booking.
3	**Realizing Smart Charging of Electric Vehicles at Public Charging Infrastructures Architecture (2024)**	Standards-based communication architecture using ISO 15118 and real-world implementation at a public charging hub.	Smart charging accuracy, communication reliability, charging control performance. Successfully enabled smart charging using open communication.	Does not provide reservation, route planning, prepaid booking, or charger reliability prediction. Further improvements to communication standards are required.

SI: NO	PAPER TITLE AND YEAR	METHODOLOGY	EVALUATION METRICS ACCURACY REPORTED	GAPS IDENTIFIED
1	**Artificial Intelligence-Based Electric Vehicle Smart Charging System in Malaysia (2024)**	AI-based smart charging using Machine Learning, Predictive Analytics, Demand Response, Dynamic Pricing, V2G integration, Mathematical optimization model.	Energy Savings, Cost Reduction, Charging Efficiency. Achieved 30% energy savings and 20.38% cost reduction compared .	Limited real-world deployment, insufficient localized Malaysian datasets, , lack of large-scale AI implementation.
2	**Smart Charging Solutions for Electric Vehicle : AI-Driven Approach to Load Balancing and Grid Integration (2025) (2025)**	AI-driven Predictive Demand Forecasting, Machine Learning, Deep Learning, Dynamic Load Balancing, Demand Response, Smart Grid Integration, Blockchain-assisted secure energy management, V2G.	Peak Load Reduction, Cost Reduction, Grid Stability, Renewable Energy Utilization, 20% reduction in peak-period overload	Centralized charging systems remain vulnerable, limited scalability , insufficient energy management.
3	**An AI Driven, Reliability-Based Planning Framework for Electric Vehicles Integrated Renewable Energy Support (2025)**	Deep Learning (GRU) for EV load forecasting, Enhanced Harris Hawk Optimization (EHHO), Bus Reliability Index (BRI), PV & Battery Storage System optimization, Reliability-aware planning.	EENS, AEENS, SAIFI, SAIDI, Voltage Profile, Power Loss, Reliability Indices. Integration of PV and BSS reduced EENS by 16% (IEEE 15-bus) .	Previous works ignored time-dependent component failures, comprehensive reliability assessment, and realistic EV charging uncertainty.

SI:NO	PAPER TITLE AND YEAR	METHODOLOGY	EVALUATION METRICS ACCURACY REPORTED	GAPS IDENTIFIED
1	**How to Distribute Charging Requests of Electric Vehicles? A Reservation-Based Approach (2023)**	Developed a **reservation system** that allows EV users to book a charging station before arriving. The system was tested using real charging station data.	Reduced waiting time, improved charging station usage, and balanced charging requests among different stations.	Users must reserve in advance. It does not use AI to predict waiting time or future charging demand.
2	**Reserving Charging Decision-Making Model and Route Plan for Electric Vehicles (2018)**	Used the **K-Shortest Path Algorithm** to find the best charging station by considering traffic, distance, charging cost, and station availability.	Reduced travel time and charging cost. Helped users choose the best route and charging station.	Does not use AI or machine learning. Focuses only on route planning and charging reservation.
3	**Electric Vehicle Charging: A Business Intelligence Model (2025)**	Used **Artificial Intelligence (AI), Machine Learning** to analyze EV and charging station data. The system recommends the best charging schedule and improves charging management.	Reduced waiting time, improved charging station usage, reduced charging cost, and provided better charging schedule recommendations.	Needs a large amount of real-time data. Difficult to implement on a large scale and requires secure data sharing.

SI: NO	PAPER TITLE AND YEAR	METHODOLOGY	EVALUATION METRICS ACCURACY REPORTED	GAPS IDENTIFIED
1	**Long Trip Charging Planning of Battery Electric Vehicle Considering Vehicle Waiting Time Forecast at Fast Charging Stations (2025)**	Proposed a Mixed-Integer Dynamic Programming (MIDP) approach integrated with Monte Carlo simulation to forecast waiting times at fast charging stations and optimize long-distance EV charging plans.	Total trip time, charging waiting time, energy consumption . Achieved up to 9.6% reduction in total trip time and up to 15.1% reduction in charging waiting time compared with conventional charging planning methods.	Assumes fixed traffic and accurate forecasts; lacks real-time traffic, dynamic pricing, renewable energy integration, and large-scale validation..
2	**Adaptive Federated Learning Framework for Predicting EV Charging Stations Occupancy (2025)**	MAE, RMSE, MAPE	Outperformed conventional federated learning methods by improving forecasting accuracy while preserving data privacy.	Further testing is needed on larger charging networks with diverse real-time traffic conditions.
3	**Reliability of Electric Vehicle Charging Infrastructure: A Cross-Lingual Deep Learning Approach (2023)**	Used **BERT-based Deep Learning** and **Machine Translation** to analyze EV charging station reviews written in different languages.	**Accuracy:** 89.1% **F1-score:** 0.78 The model successfully identified charging station problems from multilingual user reviews.	The system depends on translated user reviews. It needs better support for real-time multilingual data and more real-world testing.

THANK YOU